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RESEARCH ARTICLE

Performance Comparison and Visualization of AI-Generated-Image Detection Methods

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ABSTRACT Recent advancements in artificial intelligence (AI) have revolutionized the field of image generation. This has concurrently escalated social problems and concerns related to AI image generation, underscoring the necessity for an effective AI-generated-image detection method. Therefore, numerous methods for detecting AI-generated images have been developed, but there remains a need for research comparing the effectiveness of and visualizing these detection methods. In this study, we classify AI-generated-image detection methods by the image features they use and compare their generalization performance in detecting AI-generated images of different types. We selected five AI-generated-image detection methods for performance evaluation and selected vision transformer as an additional method for comparison. We use two types of training datasets, i.e., ProGAN and latent diffusion; combine existing AI-generated-image test datasets into a diverse test dataset; and divide them into three types of generative models, i.e., generative adversarial network (GAN), diffusion, and transformer, to evaluate the comprehensive performance of the detection methods. We also analyze their detection performance on images with data augmentation, considering scenarios that make it difficult to detect AI-generated images. Grad-CAM and t-SNE are used to visualize the detection area and data distribution of each detection method. As a result, we determine that artifact-feature-based detection performs well on GAN and real images, whereas image-encoder-feature-based detection performs well on diffusion and transformer images. In summary, our research analyzes the comparative detection performance of various AI-generated-image detection methods, identifies their limitations, and suggests directions for further research.

INDEX TERMS Generative AI, AI-generated-image detection, synthetic-image detection, performance comparison, GAN, diffusion model, transformer, Grad-CAM, t-SNE.

I. INTRODUCTION

Since the emergence of artificial intelligence (AI) image generative models, numerous advancements have been made in the field. The introduction of the generative adversarial network (GAN) [1] marked the start of AI image generation, with various GAN models having been developed to create images of diverse subjects such as faces, artwork, and landscapes. Subsequently, the diffusion model [2] was introduced, enabling the generation of higher-quality images compared to those generated by the GAN model. The emergence of AI image-generation models utilizing the transformer structure [3] found in language processing models has

facilitated the creation of superior-quality images that meet the needs of users. These AI image generative models have progressed over time, heightening the seeming authenticity and naturalness of AI-generated images and rendering them increasingly indiscernible from real pictures. Therefore, as AI images gain ground, a range of social concerns have arisen as a result of the misuse of AI images. When “Théâtre D’opéra Spatial,” a painting that was awarded first prize in the digital category of the Colorado State Art Competition 2022, was revealed to be generated by Midjourney [4], an AI image generative model, concerns were raised about the extent to which AI-generated images should be regarded as creative works, if at all, and the rightful ownership of AI-generated-image copyrights [5]. On a more serious note, a social media user posted a fabricated photo of an explosion

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near the Pentagon in the United States [6], causing the U.S. stock market to drop within a few minutes. Later, it was revealed that the photo was created by AI, demonstrating the potential for AI-generated images to cause significant real-life problems.

To address ethical concerns and potential damages arising from AI-generated images, there is a need for technology that can distinguish between real and AI-generated images. Consequently, numerous studies have been conducted on AI-generated-image detection. These studies involve training a deep learning model using both real and AI-generated images to identify features unique to AI-generated images. AI-generated-image detection research has resulted in various detection methods and the use of AI-generated features for detection. However, research on the most effective methods and feature detection models to detect AI-generated images has thus far been deficient. It is also unclear which is the optimal method for discerning images generated by different types of generative models. Furthermore, there is a need for analysis of how detection methods distinguish between real images and AI-generated images.

In this study, we evaluated and compared AI-generated-image detection methods based on their generalization performance on various AI-generated images, divided into GAN-generated, diffusion-generated, and transformer-generated images. We selected five detection methods that provide a pretrained model or training code, and vision transformer (ViT) [7] as an additional detection method for comparison. For each method, we trained the model using two distinct training datasets: ProGAN-generated images and latent-diffusion-generated images. We then tested the models on separate GAN-generated, diffusion-generated, and transformer-generated image test datasets to determine which method has the strongest generalization performance on each type of generated image. By combining multiple test datasets from existing AI-generated-image detection research literature, we constructed a rich test dataset that includes a total of 23 AI-generated-image datasets and 3 real-image datasets. To gain further insight, we conducted tests of detection performance on JPEG compression quality and Gaussian blur to analyze the robustness of each method to JPEG-compressed images and Gaussian-blurred images. We also performed visualizations using Grad-CAM [8] and t-SNE [9] to see what regions the detection method detects in the image and how it classifies the image in feature space.

The main contributions of our work are as follows:

- We built various kinds of generative-model test datasets with three different generative model structures to compare existing AI-generated-image detection methods.
- We divided the AI-generated-image detection methods into three groups based on the features they use and analyzed the differences in detection performance for each group.
- We compared the robustness of AI-generated-image detection methods to image augmentation to investigate

their performance in scenarios that disrupt AI-generated-image detection.

- Our Grad-CAM analysis revealed that the presence of learned features in the region in which detection is performed is more crucial than the region itself. On the other hand, the t-SNE analysis showed how the generated-image data are distributed and how each model detects images using the artifact method and image-encoder methods.

The remaining parts of the paper are structured as follows: Section II describes the image-generation models used to create the images that constitute the test datasets used to test the detection methods and discusses related research on AI-generated-image detection focusing on performance comparison and data augmentation. Section III introduces the six detection methods used in our study, and Section IV presents the construction of the training and test datasets. Section V describes the training setting and test setting of each method, and Section VI shows the detection performance results. Section VII contains Grad-CAM and t-SNE visualizations of the detection methods. Section VIII discusses the results, and finally, Section IX concludes our study and outlines plans and recommendations for future work.

II. RELATED WORK

A. AI IMAGE GENERATIVE MODEL

An AI image generative model is a model that can generate images using either img-to-img translation, which generates a new image in a different field while maintaining the concept of the original image, or text-to-img translation, which generates a corresponding image from a text prompt that describes the desired image. AI image generative models can be classified into three different categories based on their structural design, specifically: GAN, diffusion, and transformer.

1) GAN-BASED MODEL

GAN [1] is a structured deep learning model consisting of a generator and a discriminator. The generator produces an image that tries to fool the discriminator, whereas the discriminator tries to determine whether the image is real or fake. This adversarial method by which the generator and discriminator learn from each other improves the quality of production from the generator, resulting in a better-quality image.

Since its advent, GAN has been used to develop numerous models for generating images. CycleGAN [10] transforms an image into another with a different art style or in a different domain while maintaining the concept of the original image. The model accomplishes this by introducing cycle consistency loss, which allows the conversion to proceed only long enough to turn the converted image back into the original image. Gradually, GAN models were improved to be able to generate high-resolution images. ProGAN [11] has demonstrated that it is more effective to generate high-resolution images by gradually adding layers

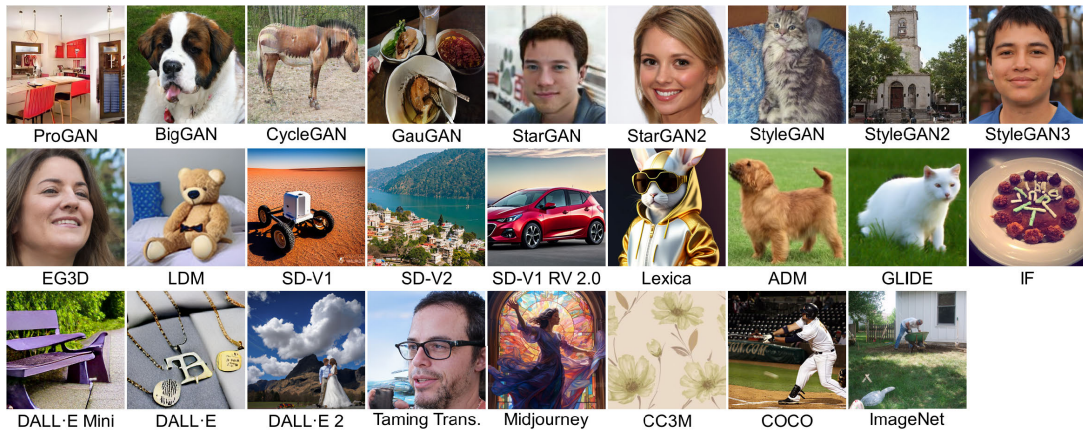


FIGURE 1. Sample images from test datasets used for our study.

to create larger images from smaller ones. On the other hand, BigGAN [12] reliably generates high-resolution images by applying orthogonal regularization to the generator with two to four times more parameters than that for the former GAN model. A GAN model capable of generating photorealistic images from a semantic layout has already been introduced: GauGAN [13] leverages spatially adaptive normalization for increased visual fidelity.

Several GAN models for generating face images have also been constructed. StarGAN [14] was proposed with a framework that enables multidomain conversion through a single model incorporating domain classification loss to generate diverse facial styles. With the addition of a mapping network and a style encoder module, StarGANv2 [15] is then able to express various styles under a specific domain, effectively converting not only human faces but also animal faces. Meanwhile, StyleGAN [16] is designed to be a style-based generator with a mapping network that generates facial images with detailed attribution. The subsequent version, StyleGAN2 [17], enhances generator normalization to eliminate artifacts in StyleGAN-produced images and advance the model. StyleGAN3 [18] was then proposed to solve the texture-sticking problem of StyleGAN2 from a signal processing perspective. Some GAN models have advanced beyond 2D image generation and can generate 3D images. For example, Chan et al. proposed EG3D [19], a tri-plane-based 3D GAN framework that enables multi-view face image generation.

2) DIFFUSION-BASED MODEL

A diffusion model generates an image by gradually adding noise to the input image through a diffusion process to create a noisy image and then learning parameters to restore the original image from the noisy image through a reverse process. This model was introduced by Wang et al. based on a denoising diffusion probabilistic model (DDPM) [2], which has been mentioned earlier. Subsequently, the ablated diffusion model (ADM) was introduced, enhancing the structural design of the DDPM and adding classifier guidance to produce class-fidelity images.

The aforementioned diffusion models have a disadvantage in that the diffusion process is performed in pixel space, resulting in a long generation time. The latent diffusion model (LDM) [20] solves this problem by performing the diffusion process on a latent space extracted through an autoencoder, resulting in a shorter execution time. The open-source model Stable Diffusion [21], which was based on LDM, was then trained using the LAION dataset [22] and merged with the CLIP:ViT-L/14 [23] encoder to use text prompts as generation conditions. The model has since been fine-tuned and utilized in various ways. Stable Diffusion v2 [24] is an official fine-tuned model trained to produce higher-resolution images and uses CLIP:ViT-H as its encoder. Stable Diffusion v1.5 Realistic Vision v2.0 [25], also a fine-tuned model, produces photorealistic images and is available on the Civitai [26] website, where fine-tuned models of Stable Diffusion are collected. Lexica [27] is another fine-tuned model of Stable Diffusion that generates smooth and glossy art images. The GLIDE [28] model enhances the classifier guidance of ADM by utilizing classifier-free and CLIP guidance to achieve photorealism in conditional images. Subsequently, the IF [29] model, consisting of a frozen T5 [30] encoder and a cascade of one base diffusion module and two super-resolution diffusion modules, was introduced. IF is a state-of-the-art model that produces image quality exceeding those of previous diffusion-based image generative models.

3) TRANSFORMER-BASED MODEL

In the image generative domain, the transformer model converts image pixels into tokens, which it then inputs into the transformer structure. Using an attention mechanism [3] that comprehends the complex connections between the tokens, the model generates new images by predicting and generating pixel tokens. DALL-E [31] is an autoregressive model that learns transformers by processing text and image tokens as a single stream based on their structure. DALL-E outperformed several GAN models in terms of Frechet inception distance [32] and inception score [33], which are used as

metrics to evaluate the generated images. In an attempt to make an open-source version of DALL-E, the model DALL-E Mini [34], trained with fewer parameters and images, was developed. Subsequently, Ramesh et al. proposed DALL-E 2 [35], a two-stage model consisting of a prior and decoder to better understand the representation of the text encoder, producing images that understood text captions better than DALL-E does. Taming Transformers [36] is a model that transforms an image into a codebook, a set of sequences that have gone through a transformer and then processed to create a new image.

4) STRUCTURE-UNKNOWN MODEL

Midjourney [4], which is considered state-of-the-art among image generative models, currently has no publicly known model structure. Midjourney is presently a paid subscription service that provides image generation in a variety of art styles and super-resolution.

B. PERFORMANCE COMPARISON IN AI-GENERATED-IMAGE DETECTION RESEARCH STUDIES

In other AI-generated-image detection studies, the typical approach is to compare the performance of the proposed method to those of existing detection methods. Past comparisons of the performance of the detection methods selected for our research study are as follows. A research study on CNNDetection [37] evaluated its performance on eleven synthesis-model-generated images, including those from six GAN models, and compared it with those of the methods studied by Zhang et al. [38]. A study on the no-down method [39] tested its detection performance on images generated by five GAN models, four diffusion models, and three transformer models and compared it with those of other detection methods [37], [38], [40]. The research study that introduced the patch selection model (PSM) [41] combined multiple generative models by family and evaluated the performance of the proposed model on each family with those of other detection methods [37], [39], [40]. Similarly, the study that developed the deep image fingerprint (DIF) [42] tested its detection performance on images generated by seven GAN models, three diffusion models, two transformer models, and Midjourney and compared it with the performance of two rule-based methods [43], [44] and artifact-based methods [37], [39]. A research study that explored universal fake image detection (UFD) tested its performance against those of other detection methods [37], [38], [40], [45] on images generated by six GAN models, three diffusion models, one transformer model, and five other generative models.

C. DATA AUGMENTATION IN AI-GENERATED-IMAGE DETECTION

The two most prevalent data augmentations employed in AI-generated-image detection research are based on Gaussian blur and JPEG compression. These data augmentation

techniques have been utilized in multiple studies to improve the generalized detection performance of AI-generated-image detection models. Xuan et al. [46] found that applying Gaussian blur and Gaussian noise to an image dataset prior to training reduced low-level noise in the images and helped the detection model learn intrinsic features. On the other hand, Wang et al. [37] showed that training on JPEG-compressed images can generalize a model to detect GAN-model-generated images and make it more robust to detect JPEG-compressed images.

III. AI-GENERATED-IMAGE DETECTION METHODS

In this section, we present the detection methods used in our study. We selected five AI-generated-image detection methods that provide training code or pretrained model weights from the official Github repository. Then, we divided the AI-generated-image detection methods into three categories based on the image features that they use.

A. ARTIFACT-FEATURE-BASED DETECTION

Artifact-feature-based detection identifies common artifacts in AI-generated images. To create a model based on this detection method, a deep learning network is trained on both real and AI-generated images. CNNDetection [37] is a widely published study that applied this approach to the detection of AI-generated images. A pretrained ResNet50 model [47] was trained using the ProGAN training dataset [37] into a binary classifier that distinguishes real images from AI-generated images, and its detection performance was evaluated on eleven CNN-based models for image generation, including six GAN models. Furthermore, the incorporation of probabilistic data augmentation techniques such as Gaussian blur and JPEG compression led to enhanced generalization performance in detecting images generated from various generative models. Subsequently, the no-down detection study [39] applied the no-down architecture [48], which does not perform downsampling nor use pooling in the first layer of the ResNet50, to achieve higher detection performance. This architecture, prevents the suppression of artifacts and allows for further calculation of the noise residual, enabling the model to learn the more comprehensive features of an AI-generated image. The detection model utilizes both the global and local features of an image by using the patch selection module (PSM) [41], was proposed by Ju et al. This model passes the entire input image through a ResNet50 model to extract a global feature map, resizes the selected local patches using a patch selection module, and passes them through the ResNet50 model again to extract local features. The global and local features are then fused using an attention-based feature fusion module to determine whether the image is real or AI-generated using a binary classifier.

Furthermore, in addition to the convolutional neural network (CNN) model employed thus far, we utilized the ViT model to evaluate the detection performance on images generated by AI models with a transformer structure. ViT is a state-of-the-art model in image classification and applies the

self-attention mechanism used in natural language processing to images, dividing them into patches and generating sequences as input to a transformer.

B. SPECTRUM-FEATURE-BASED DETECTION

Spectrum-feature-based detection is a technique for identifying distinct patterns that appear when a set of generated images is transformed into spectrum space by a fast Fourier transform. Sergey et al. introduced the deep image fingerprint (DIF) [42], which employs the inductive bias of CNN to extract fingerprints from generated images. DIF is a rule-based method that obtains an artificial fingerprint by filtering an image through a U-Net [49] high-pass filter and classifying it as either real or AI-generated by comparing the correlation between the fingerprint and the residuals extracted from the image. The DIF model has exhibited generalized detection performance even when trained on only a small number of AI-generated images, as few as 512.

C. IMAGE-ENCODER-FEATURE-BASED DETECTION

Image-encoder-based detection uses the feature space of an image-to-text encoder such as CLIP [23], which has been trained on a large number of image-text pair datasets. Ojha et al. [50] discussed the challenge of classifying both GAN and diffusion-generated images using models that were simply trained on real and AI-generated images. Therefore, they proposed a universal fake image detector (UFD) that utilizes the feature space of CLIP:ViT-L/14, which has learned from 400 million diverse images, to detect AI-generated images from various domains. The UFD model uses the CLIP model as a backbone and learns only the added linear classification layer. This model generates a feature bank for real and AI-generated images based on training images. Then, it classifies an input image using the feature bank, assessing if the input image is closer to a real or AI-generated image by calculating the cosine distance with the features inputted into the image encoder.

IV. DATASETS

A. TRAINING DATASETS

We selected two different training datasets: the ProGAN training dataset provided in the CNNDetection study [37] to test on GAN-generated images, and the LDM training dataset provided by Corvi et al. [53] to test on diffusion-generated, transformer-generated, and Midjourney-generated images. The ProGAN training dataset comprises 364K ProGAN-generated images with LSUN images and 364K real images from the LSUN dataset, all with dimensions of 256×256 size and in PNG format. From the 364K ProGAN-generated images and 364K real images, 4K each were used as the validation set. The LDM training dataset contains 200K latent-diffusion-generated images in PNG format and 200K real images from the LSUN and COCO datasets in JPEG format. From the 200K LDM-generated images and 200K real images, 20K each were used as the validation set.

TABLE 1. Specific information about different types of test datasets used in performance comparison.

Type	Dataset	Images	Source
GAN	ProGAN	4K	[37]
	BigGAN	2K	[37]
	CycleGAN	1.3K	[37]
	GauGAN	5K	[37]
	StarGAN	2K	[37]
	StarGAN2	3K	[51]
	StyleGAN	6K	[37]
	StyleGAN2	8K	[37]
	StyleGAN	1K+6K	[37], [52]
Diffusion	EG3D	1K	[53]
	Latent Diffusion	3K+3K	[53], [50]
	Stable Diffusion-V1	1K+3K	[53], [54]
	Stable Diffusion-V2	4K	[52]
	Stable Diffusion-V1 RV 2.0	4K	[52]
	Lexica	4K	[27]
	ADM	2K+1K	[53], [50]
	GLIDE	1K+3K	[53], [50]
Transformer	IF	3K	[52]
	DALL-E Mini	1K+2.5K	[53], [42]
	DALL-E	1K	[50]
	DALL-E 2	1K+0.5K	[53], [42]
Unknown	Taming Transformers	2K	[53]
	Midjourney	5K	[52]
Real	CC3M	12.4K	[55]
	COCO	13.3K	[56]
	ImageNet	13.6K	[57]

To match the size of the LDM-generated images, we rescaled and center-cropped the real images down to dimensions of 256×256 .

B. TEST DATASETS

In our study, the test dataset for a single generative model was not composed of a 50–50 split between real and AI-generated images. Instead, the real-image dataset consisted exclusively of real images, whereas the AI-generated image datasets consisted exclusively of AI-generated images. To demonstrate why, if we suppose that the Stable Diffusion test dataset has a 1:1 ratio of real and AI-generated images, it will be difficult to determine whether the detection accuracy is, for example, 100% for real images and 50% for Stable-Diffusion-generated images, or 80% for real images and 70% for Stable-Diffusion-generated images, if the detection accuracy is 75% for the whole dataset. By organizing the test dataset in this manner, we could more accurately identify the detection performance of the detection model on real images and on each of the AI-generated images. A sampling of the test images used in our study is shown in Fig. 1, and specific details of the test datasets are provided in Table 1. The “+” sign in the Images column of Table 1 signifies that the different test datasets under the Source column for that row are combined.

Our test dataset consisted of 23 AI-generated-image datasets and 3 real-image datasets. The AI-generated image datasets included images generated by 10 GAN models, 8 diffusion models, 4 transformer models, and Midjourney. All images in the test dataset were in PNG format, except for the real images and Lexica [27], which were in JPEG format.

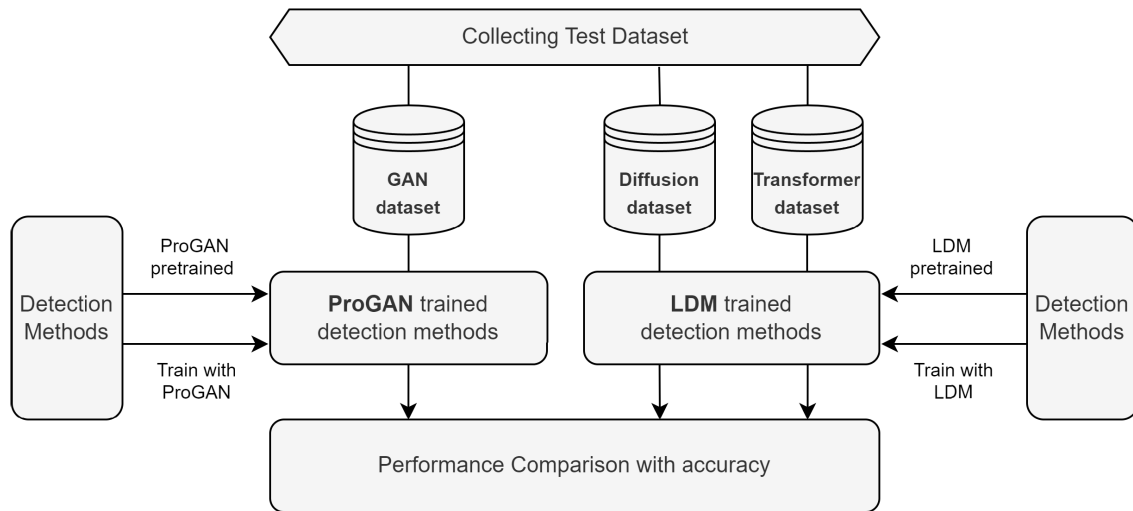


FIGURE 2. Framework for performance comparison of AI-generated-image detection methods.

Depending on the structure of the generative model, the test dataset was divided into three parts.

1) GAN-GENERATED-IMAGE TEST DATASET

The GAN-generated-image test dataset consisted of 39.3K images generated by ProGAN [11], BigGAN [12], CycleGAN [10], GauGAN [13], StarGAN [14], StarGAN2 [15], StyleGAN [16], StyleGAN2 [17], StyleGAN3 [18], and EG3D [19]. The dataset was based on the test dataset used in the CNDetection study [37]. We also used test images generated by StarGAN2, StyleGAN, StyleGAN3, and EG3D from other detection studies [51], [52], [53] for the supplementary GAN-generated images to evaluate the performance of the models against those of the most recent GAN models.

2) DIFFUSION-GENERATED-IMAGE TEST DATASET

The diffusion-generated-image test dataset contained 32K images generated by LDM [20], Stable Diffusion-V1 (SD-V1) [21], Stable Diffusion-V2 (SD-V2) [24], Stable Diffusion-V1 Realistic Vision V2.0 (SD-V1 RV 2.0) [25], Lexica [27], ADM [58], GLIDE [28], and IF [29]. This dataset was based on the test datasets used by Corvi et al. [53] and Lu et al. [52], albeit expanded with images provided in other studies on various detection methods [50], [51] and the DiffusionDB dataset [54] to increase the number of images and vary the image domains. Lexica images were obtained from the official website via crawling [27]. We conducted experiments on several Stable-Diffusion-based fine-tuned model datasets to evaluate whether detection models trained on LDM-generated images can also properly detect images generated by a fine-tuned Stable Diffusion model that has the same structure as that of the LDM.

3) TRANSFORMER-GENERATED-IMAGE TEST DATASET

The test dataset provided by Corvi et al. [53] was also used as the base of the transformer-generated-image test dataset,

which consisted of images generated by DALL-E Mini [34], DALL-E 2 [35], and Taming Transformers [36]. We used additional images from DALL-E Mini and DALL-E 2 provided in the DIF research study [42], and images from DALL-E [31] provided in the UFD study [50] test dataset. In addition to transformer-generated images, we also included Midjourney-generated images in the transformer test dataset, for a total of 13K images.

4) REAL-IMAGE TEST DATASET

The real-image test dataset is composed of the CC3M [55] validation dataset, COCO [56] test dataset, and ImageNet [57] test dataset, comprising a total of 39.3K images. The CC3M, COCO, and ImageNet datasets are widely used image datasets consisting of images from a variety of domains. When the real dataset is used with one of the aforementioned AI-generated-image datasets (GAN, diffusion, or transformer), the number of real images is adjusted to match the total number of images in the AI-generated-image dataset, resulting in a 1:1 ratio. Thus, the GAN-generated-image test dataset was paired with a real-image test dataset comprising 12.4K CC3M, 13.3K COCO, and 13.6K ImageNet images. The diffusion-generated-image test dataset was paired with a real-image test dataset comprising 10.7K images from each of the CC3M, COCO, and ImageNet datasets. The transformer-generated-image test dataset was paired with a real-image test dataset comprising 4.3K images from each of the CC3M, COCO, and ImageNet datasets. This is the most effective way to determine the performance of the detection method, because the total accuracy will be 50% even if the detection model classifies all images as real or as AI-generated.

V. COMPARISON SETTINGS

In this section, we describe the comparison-experiment settings for each method. The performance comparison framework of our study is illustrated in Fig. 2. Each detection

method was obtained from the official Github repository. If a pretrained model that was trained on the same dataset we used for training was available in the official repository, we selected it for the test. For CNNDetection and UFD, we applied the ProGAN pretrained model. CNNDetection provided several ProGAN models pretrained on images with a 50% probability of augmentation by Gaussian blur and images with a 50% probability of augmentation JPEG compression, and a model pretrained on images with a 10% probability of augmentation. We selected the model pretrained on images with a 10% probability because it performed better in the CNNDetection study [37]. For the no-down detection method, we used both the ProGAN pretrained model and LDM pretrained model. The methods without a pretrained model were trained on the ProGAN training dataset and LDM training dataset, respectively; the model trained on ProGAN-generated images was tested on GAN-generated images, and the model trained on LDM-generated images was tested on diffusion and transformer-generated images. Then, we analyzed the performance against each test dataset, using accuracy as the metric.

A. TRAINING SETTINGS

The goal of our study is to analyze the extent to which each method can achieve its highest generalization performance. Therefore, for optimal training, we set most of the training parameters to their default values in the training code rather than using the same parameters among the different methods. However, we applied the same data augmentation parameter, which significantly affects performance. The CNNDetection, PSM, and UFD models, which accounted for Gaussian blur and JPEG compression as data augmentations, were set to use the same Gaussian blur and JPEG compression probabilities of 10%, Gaussian blur sigma value range of 0 to 3, and JPEG compression quality range of 30 to 100 when training. For CNNDetection [37] and UFD [50], which provided multiple base model architectures, we used the model architectures that were used in their respective introductory studies. We used ResNet50 for CNNDetection, and ViT-L/14 for UFD. For the ViT, we used an ImageNet-pretrained ViT-B/16 model, which has a 16×16 patch size, and added the classification head to the fine-tuning configuration as a binary classifier for detecting real and AI-generated images. More detailed training settings for the detection methods are provided in Appendix A.

B. TEST SETTING

When conducting our tests, we used mostly the default values from the existing test codes of the detection methods but applied the same batch size of 1 to all methods. Although UFD was capable of determining the optimal threshold that provides the highest accuracy for a given dataset, we used a threshold of 0.5 to be fair to the other methods. The detailed test settings for all detection methods are provided in Appendix B.

In a real-world scenario, repeated saving and downloading of AI-generated images on social media can cause image quality degradation through JPEG compression. This makes it more difficult for detection models to recognize AI-generated images. Therefore, we conducted additional tests to analyze the robustness of each detection method to JPEG compression. We compressed test dataset images using the Python Imaging Library (PIL) [59] with JPEG qualities of 50, 70, and 90. In addition, a hostile adversary could manipulate an AI-generated image to evade detection. We considered a scenario where an adversary applies Gaussian blur to AI-generated images. Therefore, we applied Gaussian blur using the OpenCV library [60] to the test datasets with sigma values of 1, 2, and 3, respectively, and then evaluated the performance of the detection methods. With regard to the models for testing JPEG-compressed and Gaussian-blurred images, as described earlier, we tested the methods involving JPEG compression and Gaussian blur as augmentation techniques using a model trained with image augmentation and the other methods, ViT and DIF, using a model trained without image augmentation.

C. EVALUATION METRIC

We used accuracy as our evaluation metric for the test. The detection results of each detection method have four cases: true positive (TP), when a real image is detected as a real image; false negative (FN), when a real image is detected as an AI-generated image; true negative (TN), when an AI-generated image is detected as an AI-generated image; and false positive (FP), when an AI-generated image is detected as a real image. Equation (1) is used to calculate the accuracy. However, as explained in Section IV, the AI-generated-image dataset contains only AI-generated images, and thus, there will be no TP or FN in the evaluation results for the AI-generated-image dataset; therefore, the accuracy of detection on the AI-generated images is equal to (2). Similarly, the real-image dataset consists only of real images, and thus, there will be no TN and FP in its evaluation results; therefore, the accuracy of detection on the real images can be calculated as in (3). The total accuracy is calculated as shown in (1), which reflects the total number of AI-generated and real images, for either the GAN, diffusion, or transformer test datasets.

$$Accuracy = \frac{TP + TN}{TP + FN + TN + FP} \quad (1)$$

$$Generated\ Image\ Accuracy = \frac{TN}{TN + FP} \quad (2)$$

$$Real\ Image\ Accuracy = \frac{TP}{TP + FN} \quad (3)$$

VI. COMPARISON RESULTS

The comparison results for the detection performance of the selected detection methods are shown in Table 2, 3, and 4. The accuracy of a detection model on the AI-generated image dataset is the percentage of images detected by the model as AI-generated, whereas the accuracy of the detection

TABLE 2. GAN-generated-image and real-image detection accuracies of detection methods trained on ProGAN training dataset.

Type	Dataset	Method						Average Acc.
		Artifact				Spectrum	Image Encoder	
		CD [37]	Nd [39]	PSM [41]	ViT [7]	DIF [42]	UFD [50]	
GAN	ProGAN	100.0	100.0	100.0	100.0	89.23	99.85	98.18
	BigGAN	48.35	94.80	77.35	57.90	48.75	91.95	69.85
	CycleGAN	83.04	87.59	89.70	82.68	61.39	98.03	83.74
	GauGAN	69.90	90.16	69.60	50.28	82.54	99.70	77.03
	StarGAN	91.00	99.80	94.80	93.45	68.78	98.15	91.00
	StarGAN2	96.10	99.70	97.17	88.83	76.60	91.10	91.58
	StyleGAN	80.37	98.83	77.97	59.41	54.90	70.21	73.62
	StyleGAN2	73.86	85.81	69.29	66.70	48.71	50.18	65.76
	StyleGAN3	72.03	68.61	71.76	40.63	62.46	50.34	60.97
	EG3D	74.40	99.00	65.70	63.64	99.30	70.50	78.76
Real	CC3M	99.52	99.93	99.31	97.23	82.25	98.86	96.24
	COCO	97.56	99.83	98.33	95.32	27.71	97.49	86.27
	ImageNet	96.19	99.42	97.29	94.53	28.02	99.19	85.49
GAN Accuracy		78.27	89.36	78.64	65.30	65.39	74.41	-
Real Accuracy		97.70	99.72	98.28	95.65	45.01	98.49	
Total Accuracy		87.99	94.54	88.46	80.47	55.20	86.45	

model on the real-image dataset is the percentage of all images that the model correctly detected as real. The total accuracy is the number of correctly detected images divided by the total image dataset size. In the tables, “CD” denotes CNNDetection, “Nd” denotes no-down, all accuracies are rounded to the second digit after the decimal point, and the highest values are bolded.

A. RESULTS ON GAN-GENERATED IMAGES

The detection performance of each method trained with the ProGAN dataset on the GAN-generated images is reported in Table 2. First, the accuracies on the ProGAN images, which were generated by the same models used to generate the images on which the detection models were trained, were all greater than 99% for all tested methods except for DIF. This shows that most of the methods are good at detecting images generated by the GAN models that were also used to generate the images on which the detection models were trained. The no-down method performed well on most GAN-generated-image datasets that were not included in the training, exhibiting the best detection performance on the images generated by BigGAN, StarGAN, StarGAN2, StyleGAN, and StyleGAN2, and on the real images. Because all artifact detection methods demonstrated a performance greater than 90% on images generated by StarGAN, we can assume that ProGAN and StarGAN have similar artifacts. PSM performed slightly better than CNNDetection on the GAN-generated-image test dataset, but not as well as the no-down method. Among the six detection methods, ViT demonstrated the lowest detection accuracy on the GAN-generated images, suggesting that, as a detection model, it lacks generalization performance on GAN-generated images. Interestingly, DIF achieved the highest detection accuracy, at 99.30% for EG3D, but had a weak performance of less than 30% on the COCO

and ImageNet real images. The weak detection performance of DIF on real images can be assumed to have been due to the insufficient configuration of the ProGAN training dataset to separate the spectrum features of real and AI-generated images. With the exception of DIF, all five methods correctly classified the real images with accuracies greater than 95%. The images generated by CycleGAN and GauGAN were detected most accurately by the UFD model, achieving accuracies of 98.03% and 99.70%, respectively. However, UFD had a performance of approximately only 50% on images generated using models that are mainly for face images, such as StyleGAN2 and StyleGAN3. StyleGAN3 is the GAN model that resulted in the images the lowest detectability; herein, CNNDetection achieved the highest detection rate of 72.03%, which is higher than that of any other model. In total, when trained on the ProGAN training dataset, the no-down method outperformed all other methods on both GAN-generated and real images. Given that all methods have accuracies greater than 65% on the GAN-generated images, it can be concluded that a detection model trained only on ProGAN-generated images can detect other GAN-generated images to some extent and that the images generated by different GAN models share similar features.

B. RESULTS ON DIFFUSION-GENERATED IMAGES

Table 3 shows the detection performance of each method trained with the LDM training dataset on the diffusion-generated images. On the test images generated by LDM, which was also the model that generated the training images, all six methods achieved accuracies greater than 90%. ViT had the lowest accuracy, at 92.76%, whereas the no-down method achieved the highest accuracy, at 100%. With regard to the performance of each method on images generated by Stable Diffusion and by the fine-tuned Stable Diffusion

TABLE 3. Diffusion-generated-image and real-image detection accuracies of detection methods trained on LDM training dataset.

Type	Dataset	Method						Average Acc.
		Artifact				Spectrum	Image Encoder	
		CD [37]	Nd [39]	PSM [41]	ViT [7]	DIF [42]	UFD [50]	
Diffusion	Latent Diffusion	99.77	100.00	99.92	92.76	98.77	97.32	98.09
	SD-V1	98.85	99.93	98.85	42.05	91.78	98.45	88.32
	SD-V2	91.93	99.73	64.80	31.63	1.08	63.55	58.79
	SD-V1 RV 2.0	99.13	99.95	85.28	41.25	0.53	85.78	67.74
	Lexica	78.15	71.51	47.58	10.02	25.80	84.75	52.97
	ADM	32.33	9.20	27.77	23.87	1.40	89.43	30.67
	GLIDE	35.23	21.35	31.78	17.95	6.90	94.73	34.66
Real	IF	9.50	1.40	13.67	24.30	0.40	70.63	19.98
	CC3M	95.92	99.92	87.98	92.43	97.96	55.73	88.32
	COCO	99.86	100.00	99.73	96.39	99.88	92.09	97.99
	ImageNet	99.51	99.79	99.27	97.66	98.31	93.78	98.05
Diffusion Accuracy		72.97	68.81	63.66	39.15	34.45	86.66	-
Real Accuracy		98.43	99.90	95.66	95.49	98.72	80.53	
Total Accuracy		85.70	84.36	79.66	67.32	66.58	83.60	

model, the no-down method exhibited the highest accuracy, which was greater than 99%, on images generated by SD-V1, SD-V2, and SD-V1 RV 2.0, whereas UFD had the highest accuracy, at 84.75%, on images generated by Lexica. DIF demonstrated superior performance on images generated by LDM and SD-V1 images but underperformed compared to the other methods on images generated by the fine-tuned models SD-V2, SD-V1 RV 2.0, and Lexica. We notice that the detection performance of the detection methods decreases significantly on images generated by fine-tuned models such as Lexica that have a significantly different image style from that of the original Stable Diffusion model (SD-V1). UFD performed generally well across the different diffusion generative models, with accuracies of 89.43% on the ADM-generated images, 94.73% on the GLIDE-generated images, and 70.63% on the IF-generated images, compared to the other five methods, which all resulted in performance levels worse than 50% on the images generated by ADM, GLIDE, and IF. Overall, UFD achieved the highest accuracy, reaching 86.66%, on the diffusion-generated images. However, its performance on real images, at 80.53%, was slightly lower than those of the other methods, which exceeded 95%, with the no-down method exhibiting the highest accuracy, at 99.9%. The performance of the models in detecting differences between the real and AI-generated images is, to some extent, degraded, because the image-encoder features of the real and AI-generated images become less distinct when the detection model is trained on the LDM training dataset than when trained on the ProGAN training dataset. CNNDetection achieved the highest total accuracy, which was 85.7%. The diffusion-generated-image dataset that resulted in the lowest detection rate was the IF-generated-image dataset, with an average accuracy of 19.98%. A comparison between the detection accuracies of the models on images generated by GAN and diffusion revealed that all artifact methods exhibited a decrease in performance. This implies that images produced

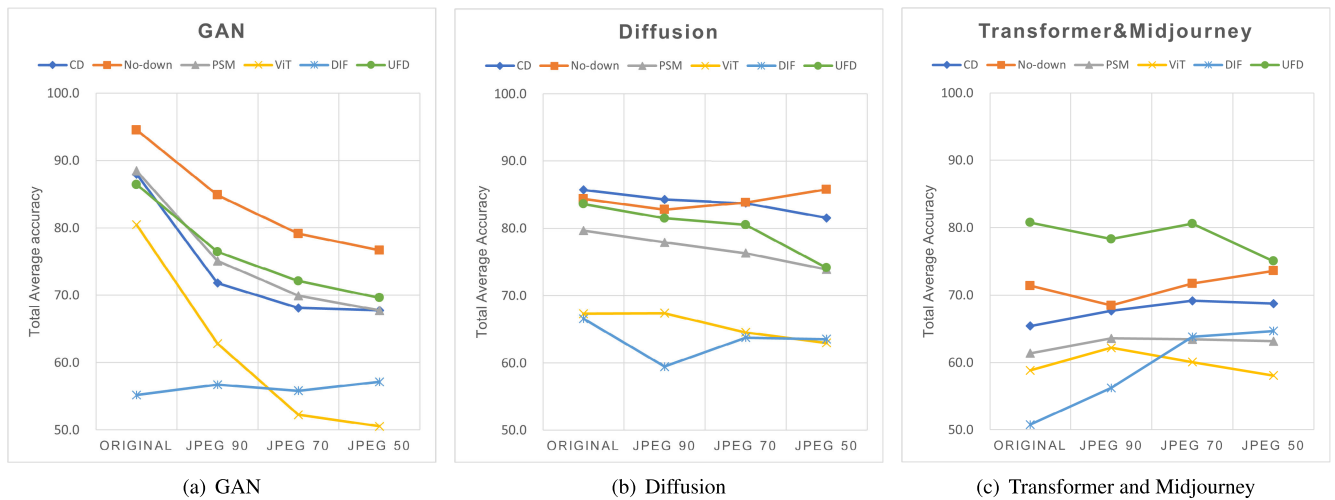
by various diffusion models have fewer shared artifact features compared to those generated by GAN models.

C. RESULTS ON TRANSFORMER-GENERATED AND MIDJOURNEY-GENERATED IMAGES

The detection performance of each method trained with the LDM dataset on the transformer-generated and Midjourney-generated images is outlined in Table 4. UFD outperformed all other methods with high detection accuracies greater than 98% on images generated by DALL-E mini, DALL-E, and Taming transformers, and 84.20% on images generated by DALL-E 2. The no-down method exhibited a performance of 71.5% and 80.2% on images generated by DALL-E mini and DALL-E, respectively, but a low detection accuracy of 2.22% on images generated by DALL-E 2. None of the detection methods successfully detected the images generated by Midjourney, which is a state-of-the-art generative model. UFD had the highest success rate, at 56.04%, whereas ViT had the lowest success rate, at 5.43%. DIF exhibited the weakest performance on images generated by transformer and Midjourney. It can be inferred that the spectrum features of diffusion-generated and transformer-generated images are quite dissimilar. The highest accuracy on the transformer-generated and Midjourney-generated images combined was 80.86% for UFD, whereas the highest real accuracy was 99.85% for the no-down method, which was similar to the results on the diffusion-generated images. Of the six methods, UFD had the highest overall accuracy. UFD showed good performance on the images generated by the transformer, which has a completely different structure from the LDM with which the training images were generated; however, its performance on real images was lower than those of the other methods. Other methods based on artifact features have shown limited detection performance on images generated by DALL-E mini, DALL-E, and Taming Transformers, but not on images generated by DALL-E 2.

TABLE 4. Transformer-generated-image, Midjourney-generated-image, and real-image detection accuracies of detection methods trained on LDM training dataset.

Type	Dataset	Method						Average Acc.
		Artifact				Spectrum	Image Encoder	
		CD [37]	Nd [39]	PSM [41]	ViT [7]	DIF [42]	UFD [50]	
Transformer	DALL-E Mini	48.40	71.50	31.93	30.39	0.37	99.71	50.38
	DALL-E	58.70	80.20	41.90	31.70	0.70	99.60	56.22
	DALL-E 2	10.36	2.22	7.26	4.10	1.75	84.20	21.16
	Taming Transformers	60.55	74.61	66.05	59.05	0.45	98.05	59.94
Unknown	Midjourney	11.50	15.00	11.36	5.43	6.24	56.04	20.03
Real	CC3M	95.75	99.86	87.63	92.43	97.92	56.29	87.49
	COCO	99.91	99.86	99.75	96.26	99.93	92.48	98.39
	ImageNet	99.52	99.82	99.33	97.65	98.27	93.16	98.02
Transformer+Midjourney Accuracy		32.48	42.92	27.19	22.27	2.82	80.86	-
Real Accuracy		98.39	99.85	95.57	95.45	98.71	80.64	
Total Accuracy		65.44	71.38	61.38	58.86	50.77	80.75	

**FIGURE 3.** Comparison of total detection accuracies on JPEG-compressed images for each test dataset.

D. RESULTS ON JPEG-COMPRESSED IMAGES

Fig. 3 illustrates the detection performance of the different methods on JPEG-compressed images.

1) GAN-GENERATED IMAGES

First, we can see from Fig. 3(a) that, for most of the detection methods working on the GAN-generated images, the detection rate decreased as the JPEG compression quality decreased. In particular, ViT exhibited the largest decrease in accuracy, with a difference of 29.93% between no JPEG compression and 50%-quality JPEG compression. DIF performed better when the JPEG compression quality was 90% than on the original images, but the overall accuracy was in the 50% range, and thus, this detection method can hardly be considered robust to JPEG compression. For the JPEG-compressed GAN-generated images, most of the tested methods exhibited decreased performance.

2) DIFFUSION-GENERATED IMAGES

From Fig. 3(b), we can see that, for the diffusion-generated images, the detection accuracies of CNNDetection, PSM, ViT, DIF, and UFD decreased as the JPEG compression

quality decreased. The detection performance of the no-down method on the JPEG-compressed images slightly increased by 1.44% at 50%-quality JPEG compression compared to its detection performance on the original images. On the other hand, UFD exhibited a detection accuracy of 74.14% on the images with 50%-quality JPEG compression, corresponding to a 9.46% loss of detection accuracy compared to on the original images, the largest decrease among the tested methods.

3) TRANSFORMER-GENERATED AND MIDJOURNEY-GENERATED IMAGES

According to Fig. 3(c), which illustrates the detection performance of the tested methods on transformer-generated images, all methods except ViT and UFD performed better when the JPEG compression quality was set to 50% than on the original images. Although the tested methods had limited detection performance on the transformer-generated images, it is feasible to conclude that the JPEG compression decreased the image quality and facilitated the recognition of comprehensive features in the AI-generated images. As a result, the no-down method emerged as the most robust to JPEG compression, with the least degradation

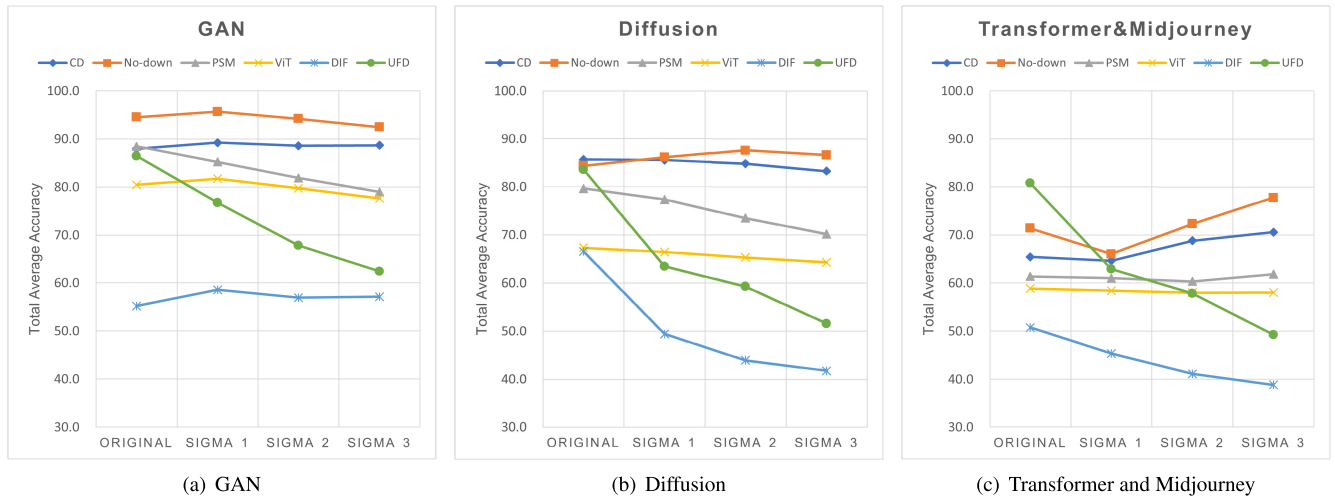


FIGURE 4. Comparison of total detection accuracies on Gaussian-blurred images for each test dataset.

in detection performance on JPEG-compressed diffusion-generated, transformer-generated, and Midjourney-generated images. CNNDetection was the next most robust to JPEG-compressed images. The same JPEG augmentations we applied to CNNDetection and PSM were also applied to UFD via training. However, UFD did not perform as well as the other two methods on the JPEG-compressed images.

E. RESULTS ON GAUSSIAN-BLURRED IMAGES

Fig. 4 illustrates the variation in total accuracy for different values of Gaussian blur sigma.

1) GAN-GENERATED IMAGES

Fig. 4(a), illustrating the result of applying Gaussian blur to the GAN-generated images, shows that the differences in the detection performance on the original and at sigma 3 were 9.48% for PSM and 24.06% for UFD, indicating significant decreases in the detection performance. By contrast, the other methods did not exhibit significant decreases in performance. The performance of CNNDetection, no-down, ViT, and DIF did not degrade much as Gaussian blur intensity was increased.

2) DIFFUSION-GENERATED IMAGES

Fig. 4(b) shows the detection results on the diffusion-generated images after Gaussian blur was applied. DIF exhibited a significant decrease in performance of 17.15% even when only a weak blur with a sigma value of 1 was applied, whereas UFD demonstrated a detection performance that was 20.14% worse at a Gaussian blur sigma value of 1 than on the original image. On the other hand, when the detection performance of the no-down method on the original images and at a Gaussian blur sigma value of 3 were compared, it was found to not only have not been reduced but, rather, increased. At a sigma value of 3, CNNDetection, PSM and ViT exhibited 2.39%, 9.49% and 3.03% lower detection performance, respectively, than on the original. For DIF, the detection performance at a

sigma value of 3 was 24.84% worse than on the original, and for UFD, it was 32.04% worse.

3) TRANSFORMER-GENERATED AND MIDJOURNEY-GENERATED IMAGES

The results for the transformer-generated and Midjourney-generated images in Fig. 4(c) show an 11.94% decrease in performance for DIF and a 31.47% decrease in performance for UFD at a sigma value of 3 compared to their performance on the original. Similar to what it exhibited on the Gaussian-blurred diffusion-generated images, UFD showed a significant drop in performance on the transformer-generated images with a weak Gaussian blur of sigma value 1. There were no noteworthy performance differences for PSM and ViT across various sigma values. On the other hand, CNNDetection showed a 5.13% better performance at sigma value 3 than on the original, whereas the no-down method showed 5.33% worse performance at sigma value 1 than on the original, but improved at sigma values 2 and 3, resulting in a 6.43% improvement in detection performance at sigma 3. Overall, increases in the Gaussian blur intensity had the least impact on the detection performance of CNNDetection and the no-down method. By contrast, UFD was found to be weak on Gaussian blurred images.

VII. VISUALIZATION

A. GRAD-CAM

To identify which parts of the image are detected as artifacts by artifact-feature detection, we performed a Grad-CAM [8] visualization of CNNDetection and the no-down method, two of the best-performing artifact-feature methods in this study. Grad-CAM is a visualization method that uses gradients to determine the weight of a layer, revealing the specific areas of an image on which the model has focused. We set the target layer of Grad-CAM to the last layer of ResNet50, which is the structure of both CNNDetection and the no-down method. Fig. 5, 6, and 7 display the Grad-CAM visualization

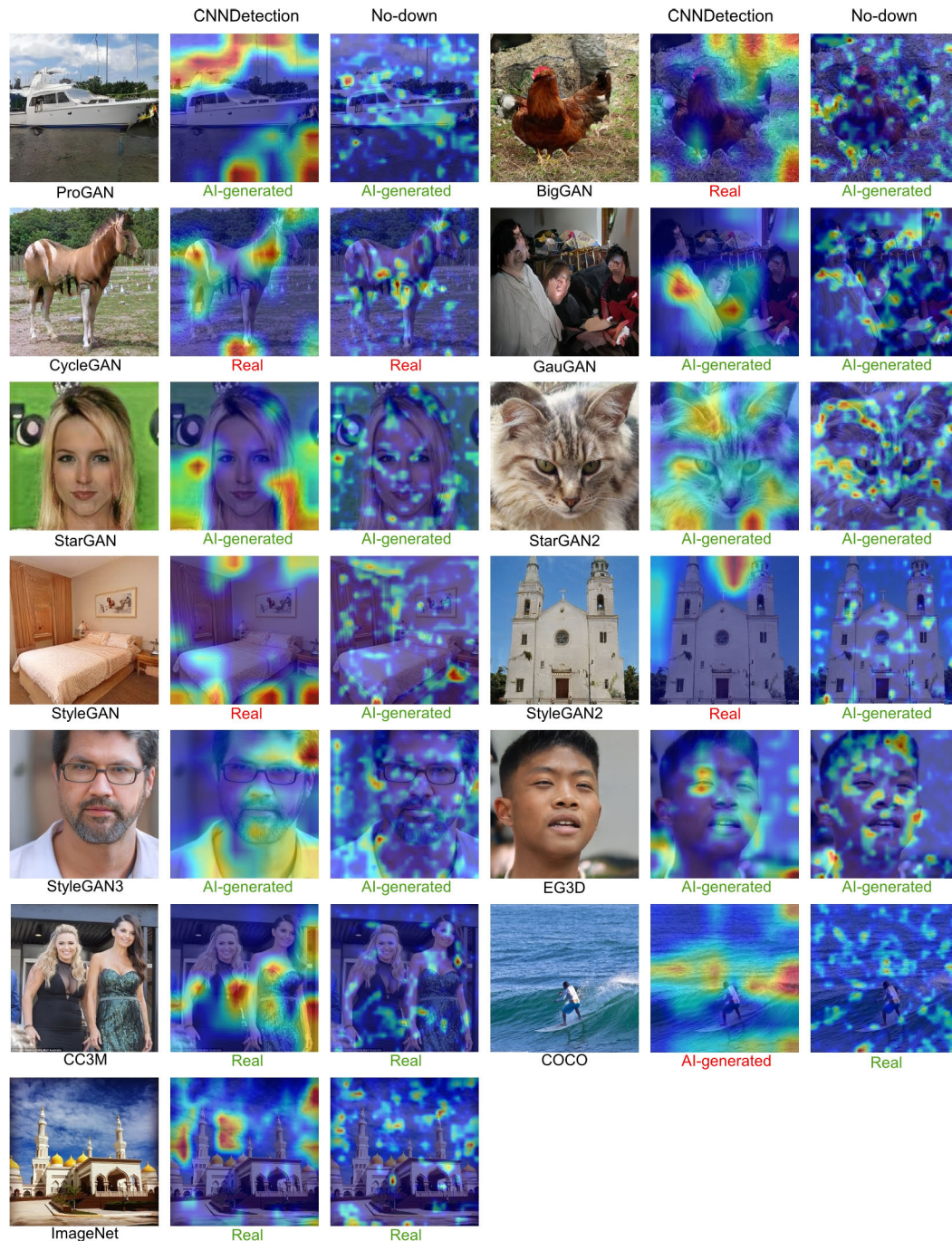


FIGURE 5. Grad-CAM visualization and detection results on GAN test dataset.

results. The first and fourth columns show an original image from each dataset, whereas the second and fifth columns show the Grad-CAM results for CNNDetection, and the third and sixth columns show the Grad-CAM results for the no-down method. Below each Grad-CAM image, we show the visualization results on how the method classified that image, coloring it green if it was classified correctly and red if it was classified incorrectly.

In Fig. 5, which shows the Grad-CAM results for the classification of GAN-generated images by CNNDetection

and the no-down method trained on the ProGAN training dataset, we can see that CNNDetection and the no-down method have different areas of focus in the image. CNNDetection concentrated on a few larger areas, whereas the no-down method concentrated on several smaller areas. Given that the no-down method subtracts a few pooling layers from its model structure, it can be assumed that this type of Grad-CAM image appears because it captures larger features than the original ResNet50 structure. In Fig. 6 and 7, the Grad-CAM results for the classification of the

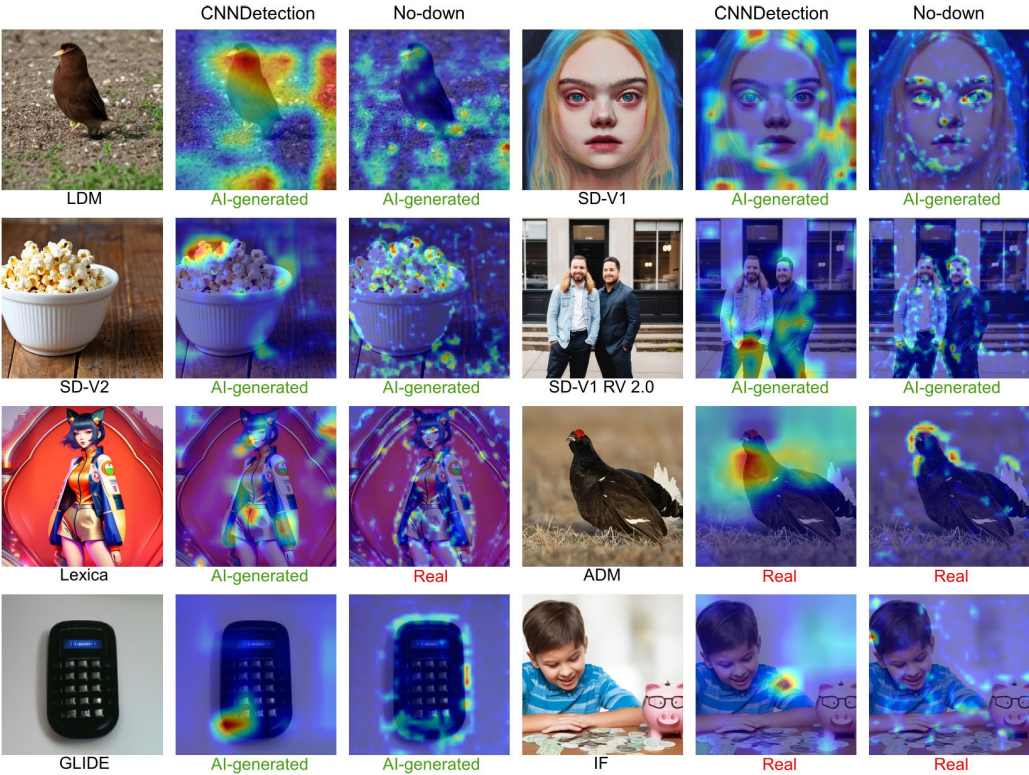


FIGURE 6. Grad-CAM visualization and detection results on diffusion test dataset.

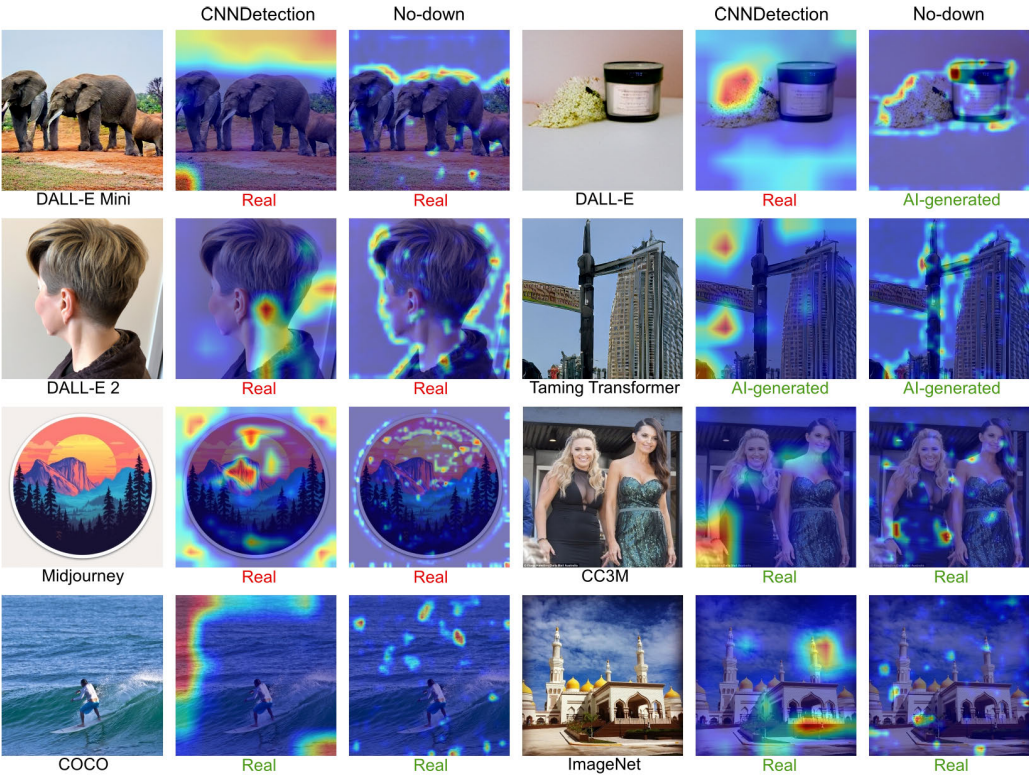


FIGURE 7. Grad-CAM visualization and detection results on transformer test dataset.

diffusion-generated, transformer-generated, and Midjourney-generated images by the two methods trained on the LDM

training dataset more clearly show the difference in the focus of the two methods. Generally, CNNDetection focused

on parts of the object in the image and background, whereas the no-down method focused along the edges of the object. Neither method showed specific difference in correct or incorrect classification based on the area of focus. On the other hand, although the images generated by the LDM and ADM are both bird images, and both methods focused on similar regions, the LDM-generated image was correctly classified, whereas the ADM-generated image was incorrectly classified. Thus, it can be concluded that the presence of artifacts in the detection regions of the AI-generated images used for training is more important than the selection of image regions by the detection method. In Fig. 5 and 7, although the original images from CC3M, COCO, and ImageNet are the same, the Grad-CAM results for the two methods are different. Furthermore, CNNDetection trained on ProGAN-generated images detected the COCO real image incorrectly as an AI-generated image, whereas CNNDetection trained on LDM-generated images detected it correctly as a real image. This demonstrates that detection results can vary depending on the training dataset.

B. T-SNE

To see how the artifact-feature method and image-encoder-feature method divide the generated images differently in feature space, we visualized them using t-SNE [9]. T-SNE is a technique for dimensionality reduction that enables the visualization of data clustering. We chose CNNDetection as the artifact-feature method and UFD as the image-encoder-feature method to visualize using t-SNE, and we extracted feature values from the previous layer of the classification layer. CNNDetection is composed of a ResNet50 structure, with a feature-value size of [2048]. By contrast, UFD is composed of a CLIP:ViT/L-14 structure, with a feature-value size of [768]. The images used for the t-SNE visualization were 600 real images and 600 AI-generated images from each of the three test datasets, i.e., GAN-generated, diffusion-generated, and transformer-generated images, for a total of 1200 images per experiment. The datasets for t-SNE visualization were constructed for each generation model structure as follows: for the 600 real images, 200 images from each of the three real-image datasets were obtained; for the GAN-generated images, 60 images from each of the 10 GAN-generated-image datasets; for the diffusion-generated images, 75 images from each of the eight diffusion-generated-image datasets; and for the transformer-generated and Midjourney-generated images, 120 images from each of the four transformer-generated-image datasets and a Midjourney-generated-image dataset. The t-SNE results for the GAN-generated images used feature values from the model trained on the ProGAN training dataset, whereas the t-SNE results for the diffusion-generated, transformer-generated, and Midjourney-generated images used feature values from the model trained on the LDM training dataset. The t-SNE results of each dataset for the CNNDetection and no-down methods are shown in Fig. 8.

First, in Fig. 8(a), which illustrates the t-SNE results for the GAN-generated images as classified by the CNNDetection and UFD methods, we can see that the cluster of ProGAN images used for training CNNDetection is located far from the cluster of real images in the feature space. We can also see that the unseen GAN-generated image has a boundary that separates it from the real images to some extent. Fig. 8(a) demonstrates that, in t-SNE, UFD is able to cluster each GAN model more effectively than CNNDetection; it can be observed that the StarGAN-generated and StarGAN2-generated images are clustered in the upper-right corner, whereas the StyleGAN3-generated and EG3D-generated are also clustered at the top. Based on the detection performance of UFD on the GAN-generated images, as outlined in Table 2, it can be concluded that clustering quality does not always correlate with detection performance. However, owing to the nature of UFD, which detects AI-generated images by comparing the cosine distance to the feature bank of real images with the cosine distance to the feature bank of AI-generated images, clustering quality tends to affect performance to some extent. In Fig. 8(b), we can notice that CNNDetection trained on the LDM-generated-image training dataset separates the LDM-generated images from the real images, whereas the images generated by other diffusion models are evenly distributed with the real images. On the other hand, the t-SNE result for UFD in Fig. 8(b) shows that the LDM-generated images are more scattered, whereas the Lexica-generated images are distinctly clustered from the real images. Fig. 8(c) illustrates that, for both CNNDetection and UFD, the Taming-Transformers-generated images are highly clustered. Furthermore, for CNN detection, the Midjourney-generated images are mixed with real images, whereas for UFD, the Midjourney-generated images are clustered mostly separately from the real images. Overall, in Fig. 8, the distribution of real images shows that CNNDetection leads to more clustering to one side than does UFD.

In summary, CNNDetection clustered well on images generated by ProGAN, LDM, and Taming Transformers, whereas UFD clustered well on images from a wider range of generative models. In addition, UFD leads to better clustering for GAN-generated images compared to for diffusion-generated and transformer-generated images. This is because GAN-generated images are often limited to a specific domain. CNNDetection is a method for detecting GAN-generated images that performs well by identifying images that share similar artifacts with the AI-generated images used for training. This is achieved by learning to widen the gap between real and AI-generated images in the feature space. On the other hand, the UFD image encoder method uses the CLIP image encoder to represent images as text features. This feature enables the detection of generated images that resemble the domain of the AI-generated images used for training. Thus, it appears that images generated by diffusion, transformer, and Midjourney, which were not included in the training, could be detected with certain generality.

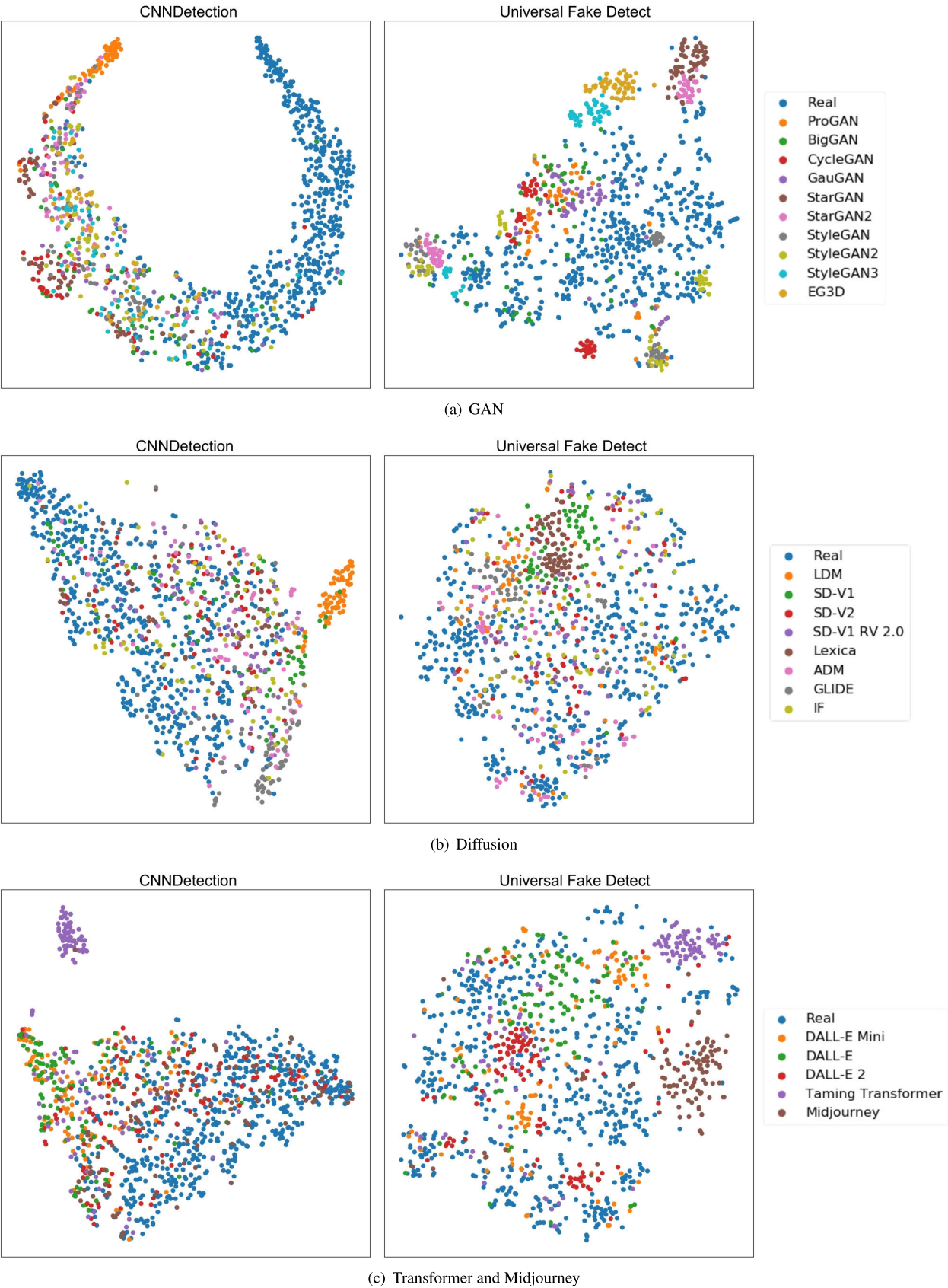


FIGURE 8. T-SNE visualization of CNNDetection and UFD methods on different generative-model image datasets.

VIII. DISCUSSION

A. AI-GENERATED-IMAGE DETECTION PERFORMANCE

With regard to performance on GAN-generated images, the detection methods based on artifact features demonstrated mostly good detection performance, of which the no-down method exhibited overwhelmingly superior detection performance with a total accuracy of 94.54%. UFD, which detects based on image-encoder features, showed good performance on images generated by various unseen GAN models, especially for CycleGAN and GauGAN, but had weak performance on images generated by StyleGAN2 and StyleGAN3. For images generated by diffusion models, CNNDetection and the no-down method performed well on images generated by latent diffusion and Stable Diffusion models, whereas UFD performed well on images generated by unseen models such as ADM, GLIDE, and IF. Excluding images generated by latent diffusion and Stable Diffusion-V1, DIF did not properly detect diffusion-generated images, classifying them as almost real images. Compared to GAN models, diffusion models generate images of various domains that are expected to share fewer spectrum features. ViT, which was used as an additional detection method in our study, did not perform as well as other methods as an AI-generated-image detection model. Because the inductive bias in the CNN structure is not present in the transformer structure [7], it is estimated that, for proper performance, it is necessary to train the model on a larger number of images than that present in the dataset we used for training the detection models.

In addition, we found that even though the detection methods were trained on LDM-generated images, the detection performance decreases significantly for images generated by fine-tuned Stable Diffusion models such as Lexica, which is essentially the same as an LDM. As more open-source image generative models are fine-tuned, detecting images generated by these models will become progressively more challenging. Therefore, the detection model should be able to capture features that are intrinsic to the generated image that do not change with fine-tuning the generative model. Artifact-based methods were mostly good at detecting images from generative models that were not significantly different from latent diffusion, which is the model that generated the images on which they were trained, but were not as good at detecting generated images from state-of-the-art generative models with significantly different structures, such as IF and Midjourney. UFD, an image-encoder-based method, performed reasonably well on diffusion-generated image datasets except on the SD-V2-generated images, where its performance exceeded 70%. In particular, UFD showed superior performance on untrained images, which exceeded 84% on all transformer-generated images. However, UFD had the limitation of performing worse than artifact methods on real images. It can be concluded that the artifact method is suitable for detecting AI-generated images that share similar features with the training images, yielding very high accuracies, whereas the image-encoder method exhibited good performance that generalizes to most generated images.

For further research on trusted AI-generated-image detection models, it is possible to create an ensemble model that combines the advantages of the artifact and image-encoder methods.

B. ROBUSTNESS TO DATA AUGMENTATION

The change in detection performance with JPEG compression and Gaussian blur varied slightly depending on the detection method and the structure of the generative model that produced the input image. DIF and ViT, which did not include JPEG compression and Gaussian blur as data augmentations during training, generally performed worse than the other methods on augmented images. As the quality of JPEG compression decreased, the detection performance on GAN-generated images mostly decreased; however, the no-down method exhibited the most consistent detection performance on diffusion-generated and transformer-generated images, followed by CNNDetection. When Gaussian blur was applied, CNNDetection and the no-down method showed the most stable performance on images generated from the three different generative-model structures. By contrast, UFD had a significant decrease in performance when detecting Gaussian-blurred images. To ensure the robustness of the detection method against attacks that attempt to evade detection using JPEG-compressed images and Gaussian blur, it is necessary to use JPEG compression and Gaussian blur as data augmentation during the training process.

C. VISUALIZATION OF DETECTION METHODS

Based on the Grad-CAM results for CNNDetection and the no-down method, we noticed that even though they were trained on the same dataset, the focus areas on the input images differed depending on the model structure. Furthermore, seeing that the detection results of the detection method are less related to the focus areas on the input images, it can be inferred that it is more important to determine the presence of artifacts in the detection areas on the AI-generated images used to train the detection models rather than the selection of focus areas that affect the detection results. With regard to the detection performance, we found that the same image can be classified correctly or incorrectly depending on the training dataset of the detection method, showing the importance of configuring the training dataset of the AI-generated-image detection method.

After that, we used t-SNE visualization to investigate the data distribution difference in feature space between artifact detection and image-encoder detection for real and AI-generated images. Because the distributions of the ProGAN and LDM images used for training were far from the distributions of the real images, CNNDetection was well able to detect images with similar artifacts among the AI-generated images that were not used for training, i.e., images with distributions close to the AI-generated images that were used for training. It signifies that the artifact method assumes that the AI-generated image being classified has similar artifacts to those of the images on which it was

TABLE 5. Training parameters for detection methods used in our study.

Parameters	Method				
	CNNDetection	Patch Selection Module	Vision Transformer	Spectrum	Image Encoder
Base Model	ResNet50	ResNet50	ViT-Base-Patch16	U-Net	CLIP: ViT-Large-Patch14
Pretrained Dataset of Base Model	ImageNet	ImageNet	ImageNet	-	400M image-text pairs
Batch Size	64	64	16	64	256
Epoch	30	30	4	60	30
Initial Learning Rate	0.0001	0.0001	0.0002	0.0005	0.0001
Optimizer	Adam	Adam	AdamW	AdamW	Adam
Image Rescale Size	256	256	224	-	256
Image Crop Size	224 center crop	224 center crop	-	256 center crop	224 center crop
Gaussian Blur Probability	10%	10%	-	-	10%
Gaussian Blur Sigma	0.0-3.0	0.0-3.0	-	-	0.0-3.0
JPEG Probability	10%	10%	-	-	10%
JPEG Compression Quality	30-100%	30-100%	-	-	30-100%

trained. This can be seen as a limitation when detecting images from new generative models. On the other hand, the t-SNE results from UFD indicate that the CLIP image encoder enables AI-generated images that were not included in the training to cluster by model in the feature space on some level. Although the degree of clustering was not always proportional to the detection performance, it was possible to demonstrate some degree of generalized detection performance across generative models. A higher-performing detector can probably be created if CLIP can exhibit this level of clustering performance on AI-generated images without training on them and if a dataset with a larger number of AI-generated images can be built.

IX. CONCLUSION AND FUTURE WORK

In our study, we compared the detection performance of six AI-generated-image detection methods on a total of 23 AI-generated-image datasets consisting of GAN-generated, diffusion-generated, transformer-generated, and Midjourney-generated images. We categorized the six detection methods into three groups: artifact, spectrum, and image encoder, based on the image features used for detection, and analyzed the effectiveness of each detection method in detecting AI-generated images according to their respective categories. We then tested how detection performance changed when JPEG compression and Gaussian blur were applied to the images. We also used visualization techniques to show the differences between artifact-feature-based detection methods in terms of their focus regions on an image, and between detection methods using different features in terms of their distribution of images in feature space. Subsequently, we correlated these differences with detection performance. In summary, our analysis focused on the objective evaluation of the detection performance of and proposed future directions for AI-generated-image detection methods.

Detecting AI-generated images is increasingly challenging owing to the continual development of more image generative models that produce better and higher-quality images. Therefore, our next research study will collect images from state-of-the-art generative models, such as Stable Diffusion XL [61] and DALL-E 3 [62], and evaluate their detection

performance against additional AI-generated-image detection methods. After that, we will use the LIME [63] visualization technique to analyze the areas that the detection method focused on with more precision.

APPENDIX A TRAINING SETTING

We used a single RTX A5000 GPU as the computing environment for training our selected detection methods. The training parameters for the five detection methods, i.e., excluding the no-down method, are shown in Table 5. For the no-down method, models pretrained on ProGAN-generated images and LDM-generated images were obtained instead. We used the same parameters for training the models on the ProGAN-generated-image training dataset and the LDM-generated-image training dataset. We referred to the post [64] by Hugging Face for the code and parameters for fine-tuning ViT. As described in Section V-A, most of the parameters of each method adopted the default values from the original training code, except for the data augmentation parameter. Additionally, DIF has a parameter to set how many real and AI-generated images to use when training; its default value is 512 images. However, given that it may not perform very well when trained with fewer images than those used for the other methods, we set this parameter to 360K for the ProGAN-generated-image training dataset and 180K for the LDM-generated-image training dataset and conducted additional training and testing. In additional tests, the performance was higher when 360K and 180K images were used than when only 512 images were used. Therefore, we adjusted the parameter for the number of training samples for DIF to match the number of training dataset images used by the other methods instead of simply following the default value.

APPENDIX B TEST SETTING

After training for a set number of epochs per method, the model at the epoch with the best evaluation results on the validation set was used for the test. Given that DIF has no validation set in the training process, we tested all models every 5 epochs and then used the model with the best results. When testing the detection models on the images generated

by GAN, diffusion, transformer, and Midjourney, we all used the same test parameters for each method, which are as follows:

- **CNNDetection**

Batch size 1, No resize, No crop, No JPEG compression, No Gaussian blur.

- **No-down**

Batch size 1, No resize, No crop, No JPEG compression, No Gaussian blur.

- **Patch Selection Module**

Batch size 1, No resize, No crop, No JPEG compression, No Gaussian blur.

- **Vision Transformer**

Batch size 1, 224×224 resize (Interpolation Mode: Bilinear), No crop.

- **Deep Image Fingerprint**

Batch size 1, No resize, 256×256 center crop.

- **Universal Fake Detect**

Batch size 1, No resize, 224×224 center crop, No JPEG compression, No Gaussian blur.

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